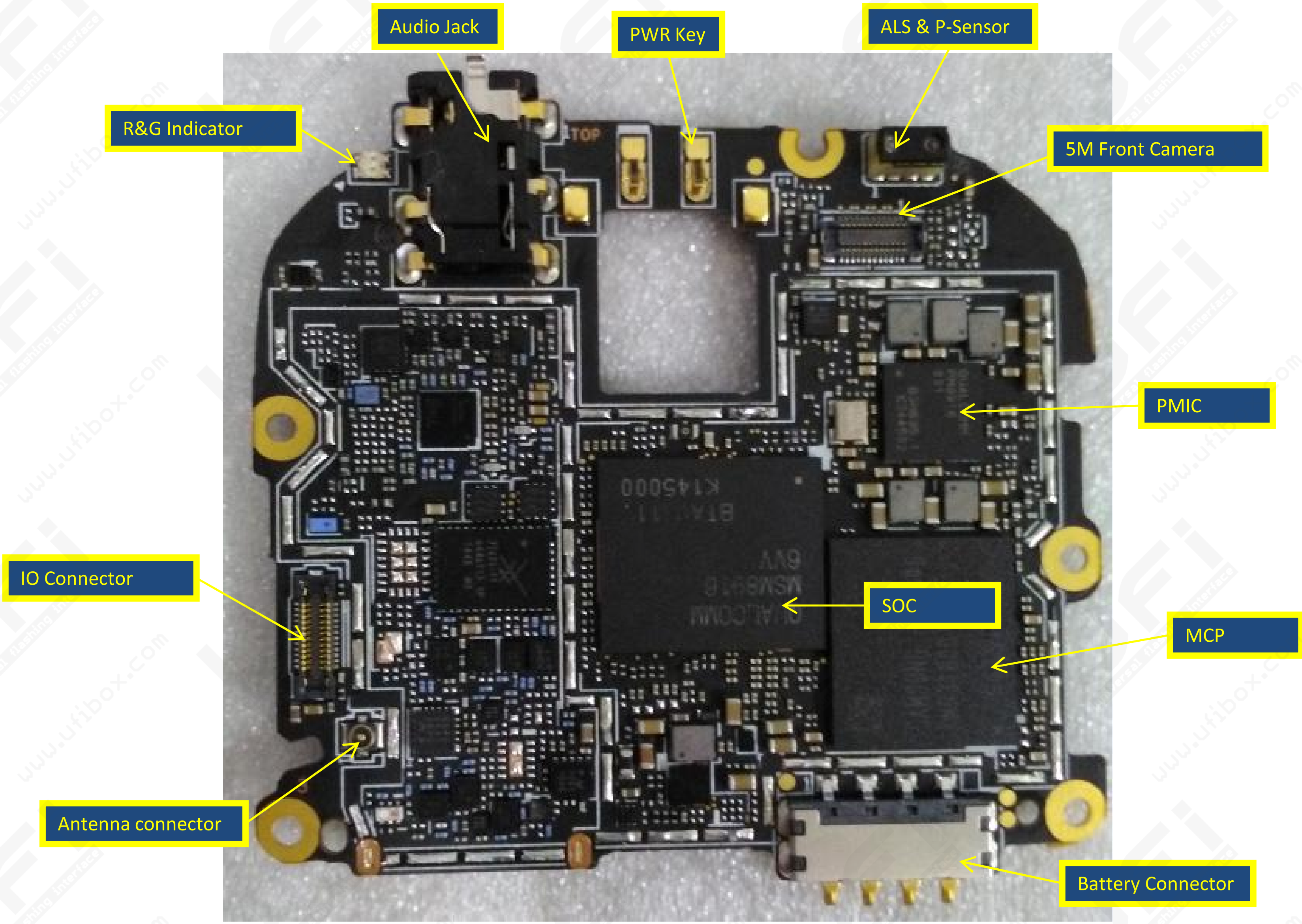
ZE500KL

Hardware Introduction

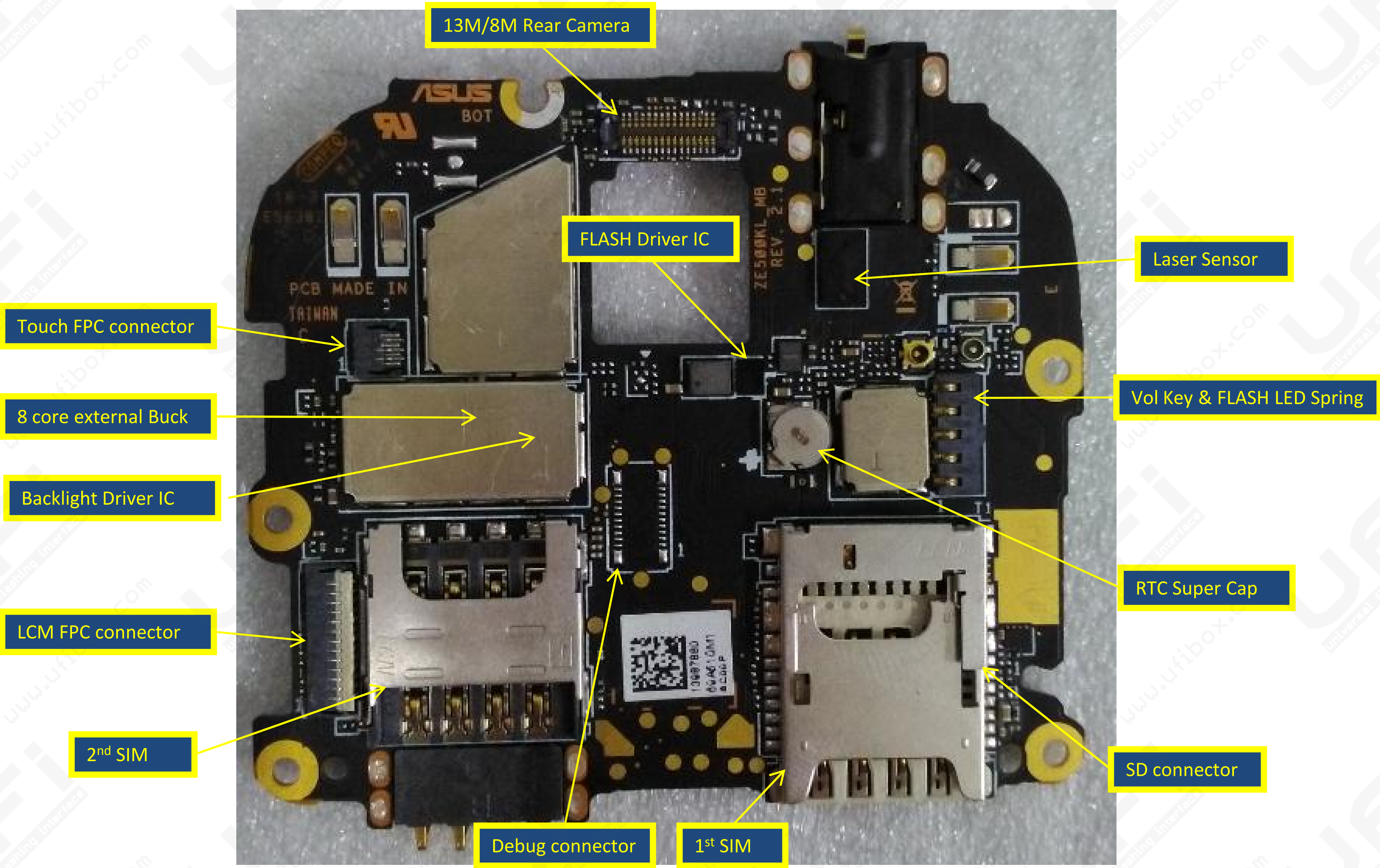
2015.3.24

System Introduction

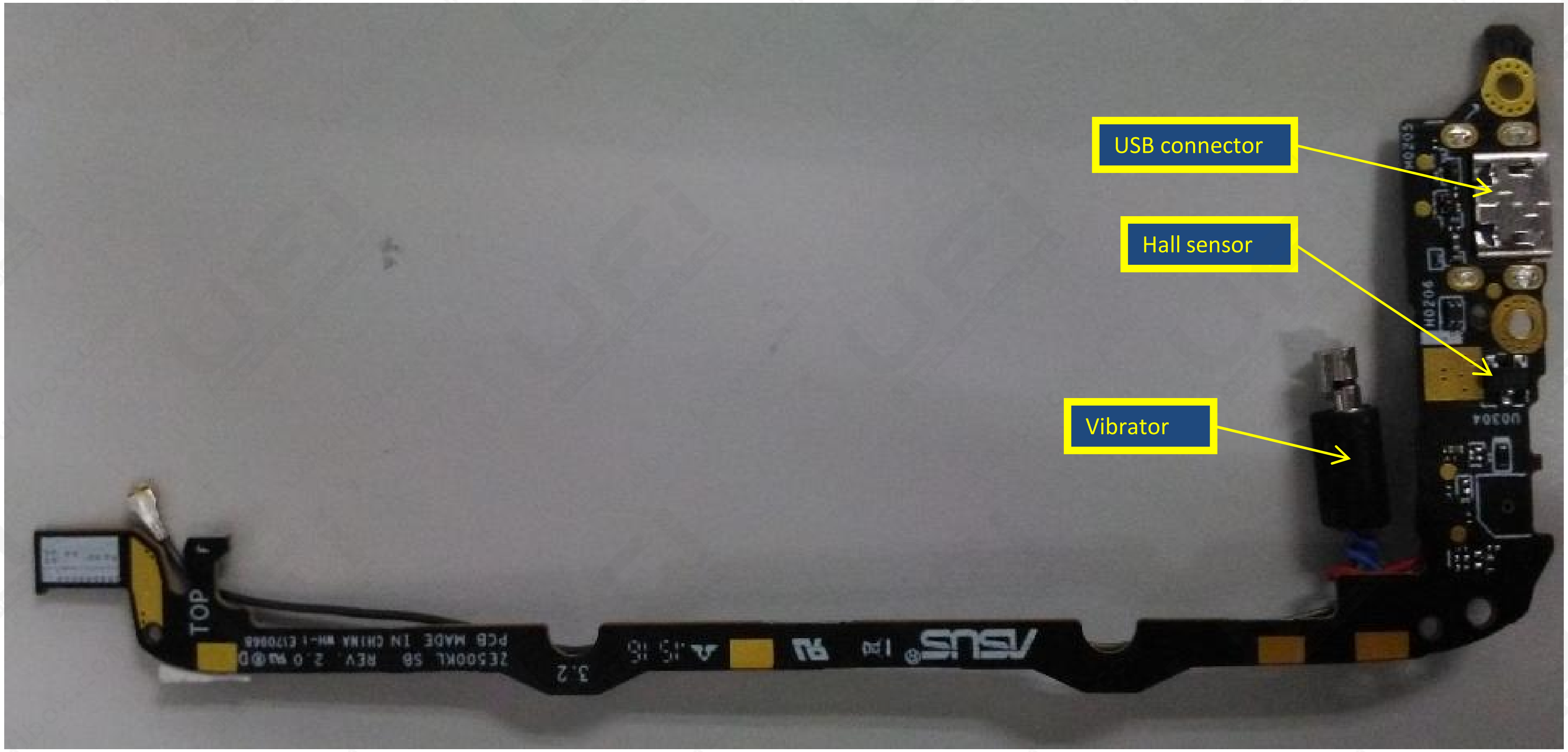
Assembly Chart TOP VIEW



Assembly Chart BOTTOM VIEW

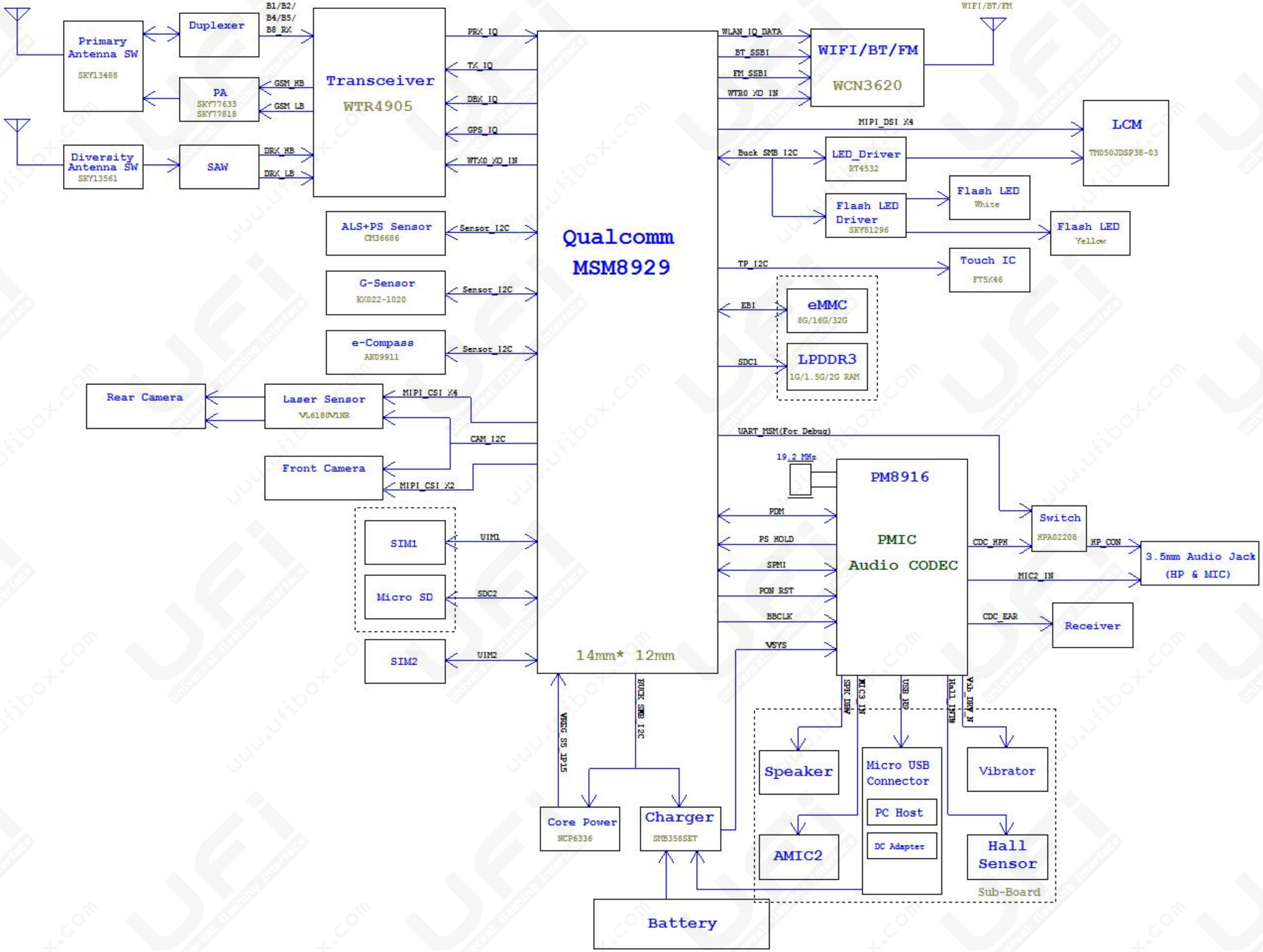


Assembly Chart TOP VIEW



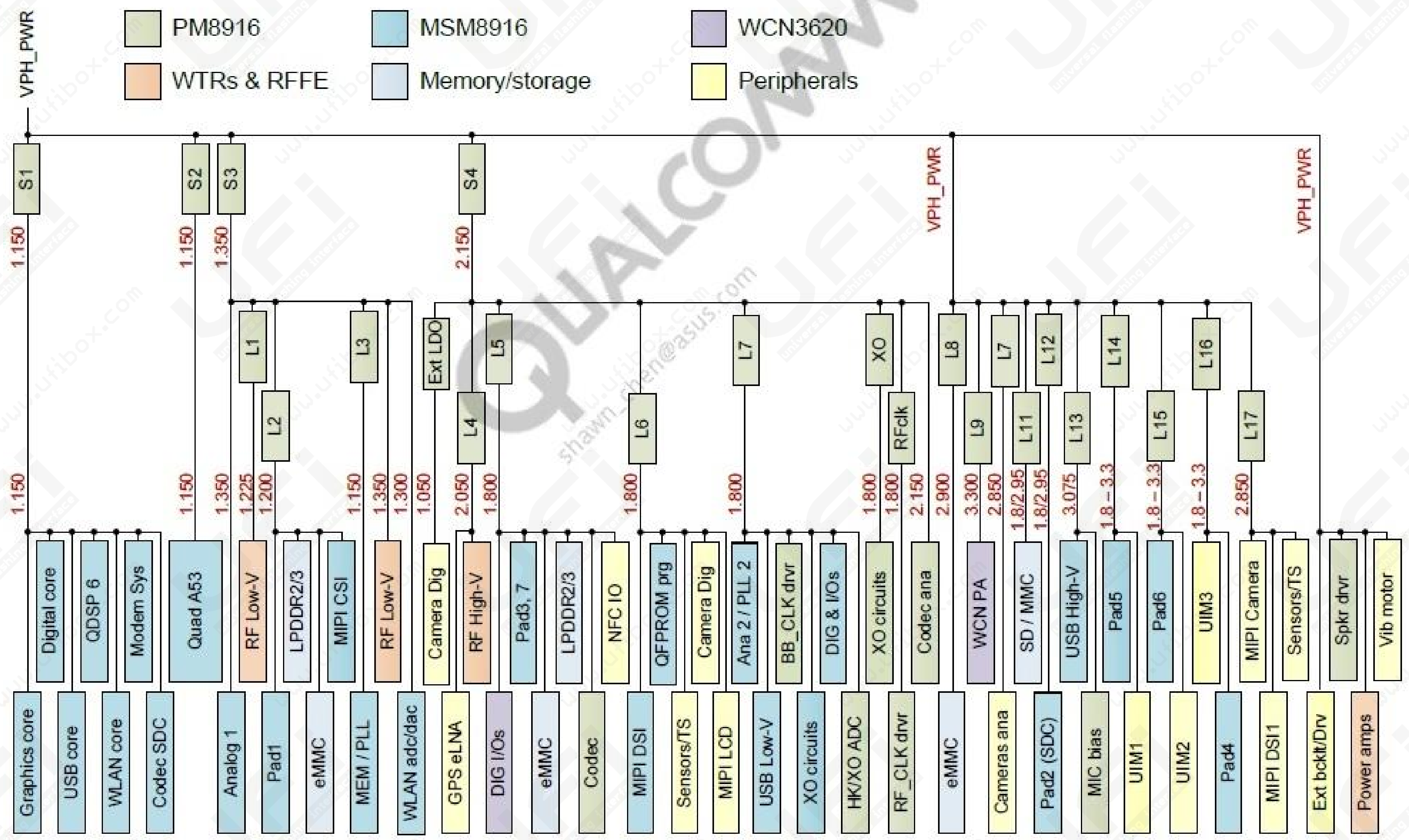
Assembly Chart

System Block Diagram



System Power Architecture

MSM Power Grid



POWER ON SEQUENCE

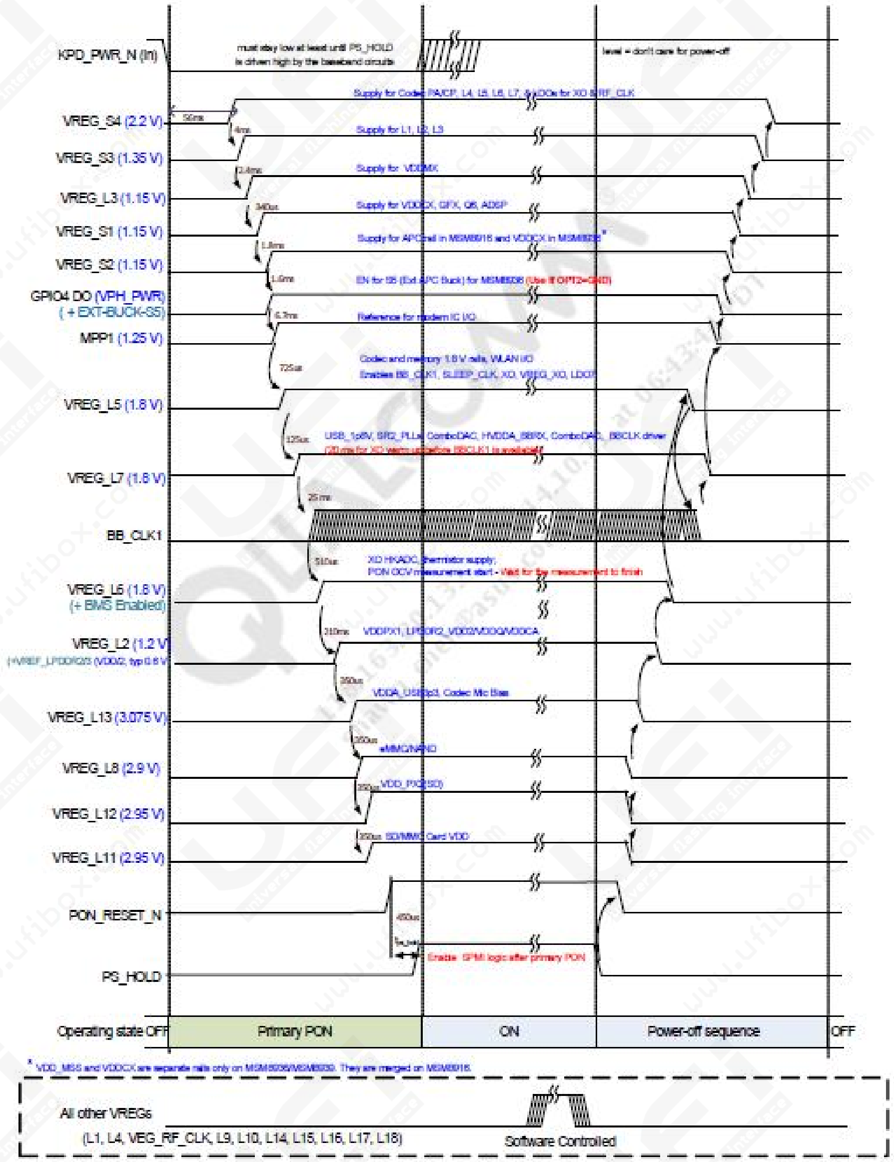


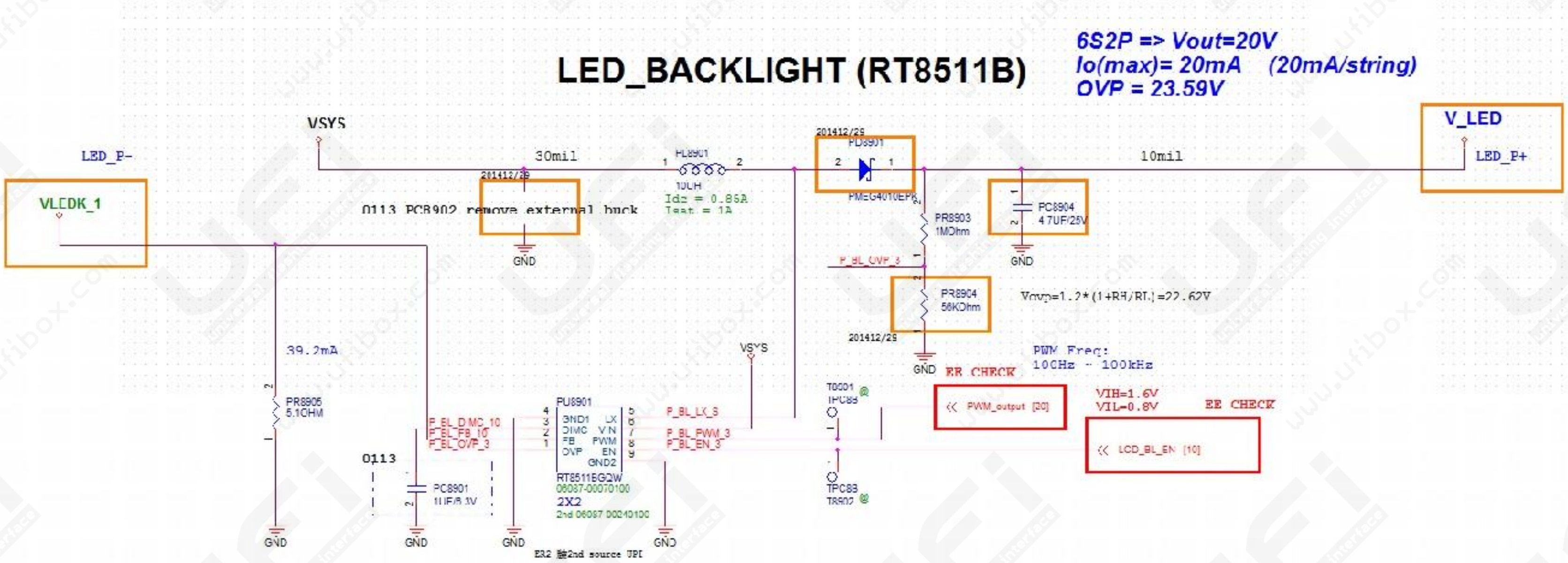
Figure 3-8 Poweron sequence

Trouble shooting

1. No display
2. No Boot
3. Touch error
4. SD card error
5. P-Sensor/Light Sensor error
6. Camera error

No display

1. To confirm FPC has connected
2. Exchange Panel, check Panel is normal
3. When the boot, measure V_LED then the voltage is greater than 5V
4. Check LCD_BL_EN is high, PWM output has SPWM Input
5. To confirm PR8905 is 5.1 ohm
6. To confirm there is no damage PL8901 with PD8901



No Boot

1. ZE500KL not supported on automatic AC power, only charging mode
2. Confirm that the battery voltage is greater than 3.6V
3. To confirm Panel FPC has connected
4. To confirm KYPD_PWR_N pull low when you press Power Button
5. To confirm PS_Hold pull high (1.8V)
6. Check power(power on sequence) to-ground has no short circuit
7. Measure power(power on sequence), all power must have it
8. Please confirm XTAL must work

Other system status

- ## When boot, USB is not recognized system

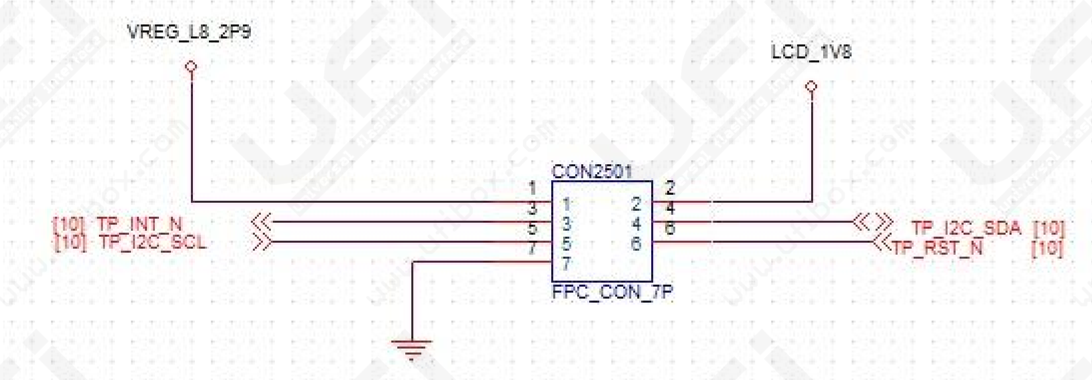
1. Check the sub board has buckled , C-clip is clamped on
2. Check if the sub board has bent/ Broken status

- ## The phone can not charge

1. It will be charged when the AC comes in, not need control signal. Check the adapter has power and battery has plugged in, then the appearance of sub board is ok
2. Replace the PU 8101.

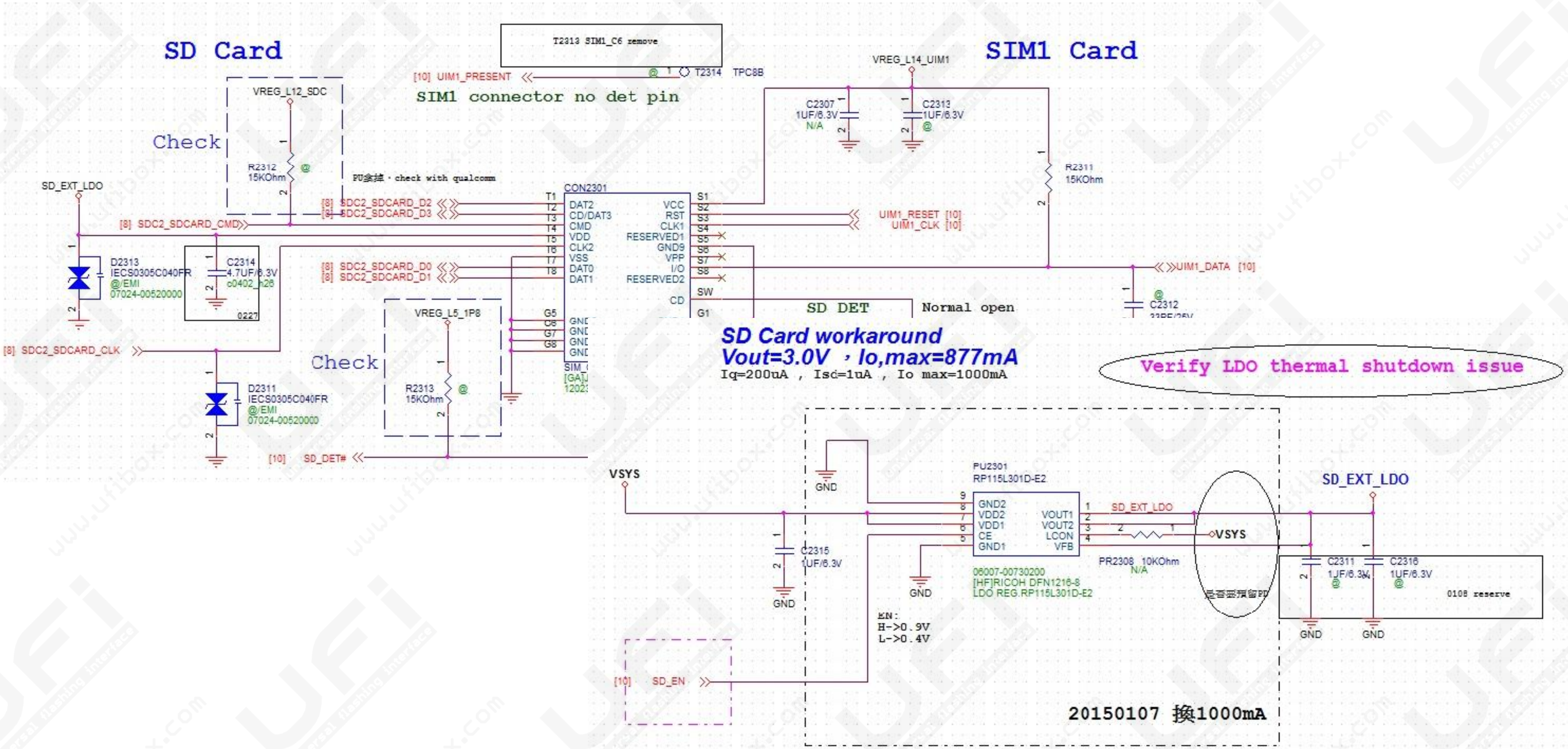
Touch error

1. To confirm PC has not connected
2. Exchange Panel, To confirm TP is normal
3. To confirm has no voltage LCD_1V8 and VREG_L8_2P9
4. Check the I2C have no PU
5. Check TP_RST_N is high
6. When touch the screen, TP_INT_N has no high-low change



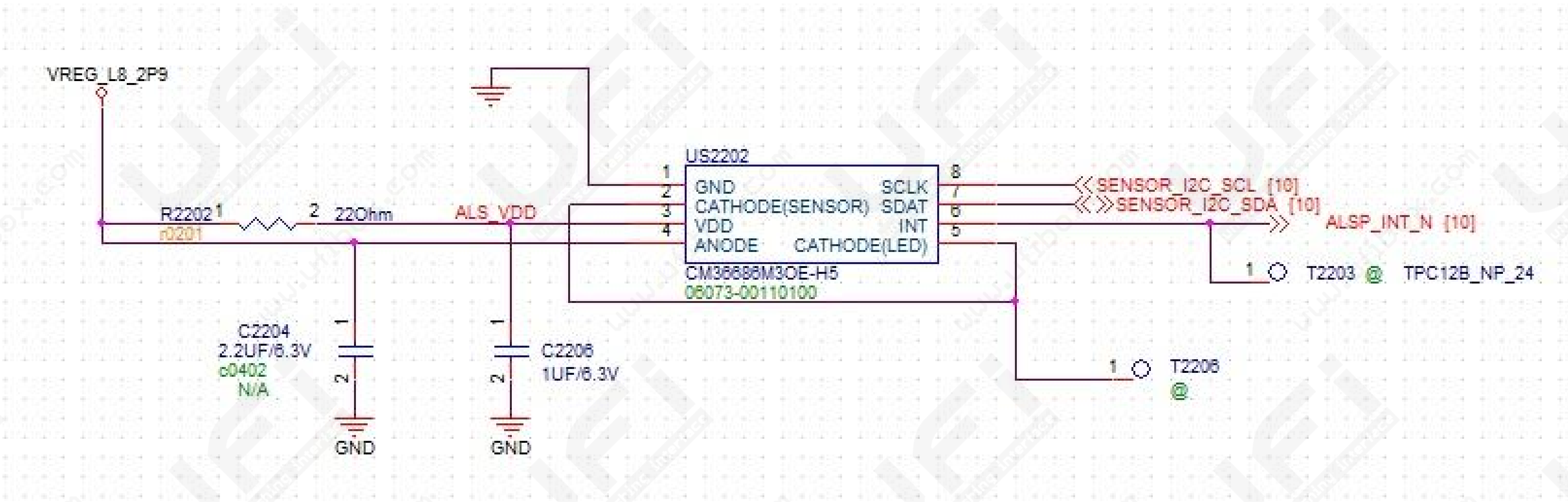
SD card error

1. Please confirm when you insert SD card, SD_DET # will pull low
2. Confirm SD_EXT_LDO has 3V output(If not, check the SD_EN is high)
3. Has 3V output but turn off, check DATA0 ~ 3 / CMD / CLK is disconnected



P-Sensor/Light Sensor error

1. Reassemble Rubber and retest
2. Check I2C is ok
3. Check VREG_L8_2P9 has Voltage



Camera error

1. Replace the Camera module and check module is normal
2. Open Camera, check FCAM_1V2,FCAM_AVDD_2V8 and FCAM_DVDD_1V8 have power